

Dark Pattern Detector

Detecting Manipulative Text on E-Commerce Websites using NLP

Submitted by: Abhishek Joshi & Shaurya Ghorpade

Course: Artificial Intelligence | Date: 30 September 2026

Abstract. Dark patterns are pieces of interface text designed to pressure online shoppers, such as fake countdown timers and false stock warnings. We fine-tuned a DistilBERT transformer on 2,356 labelled e-commerce texts and compared it with TF-IDF baselines: it reached 96.83% F1 on the test set. Because the training data comes from Western websites, the model misjudged Indian shopping text, so we adapted it with Indian-style examples, raising accuracy on a held-out Indian set from 65.63% to 96.88%. The final model powers a web dashboard that scans any product page link and shows where and how the page pressures the shopper.

1. Introduction and Problem Statement

Dark patterns are user-interface design tricks that push people into decisions they did not intend to make. On shopping websites they include fake countdown timers ("Deal ends in 04:59"), false scarcity ("Only 2 left in stock!"), pressure from other shoppers ("27 people bought this in the last hour") and guilt-tripping opt-out buttons ("No thanks, I don't like saving money"). A crawl of about 11,000 shopping websites by Mathur et al. (2019) found 1,818 instances. In India, the Central Consumer Protection Authority (CCPA) issued the **Guidelines for Prevention and Regulation of Dark Patterns, 2023**, which ban 13 such practices, including false urgency and confirm shaming. Checking millions of product pages by hand is impossible, so regulators and consumers need automatic detection.

Objectives: (1) build and compare NLP models that classify website text as a dark pattern or not; (2) identify the type of dark pattern; (3) make the model work on Indian shopping sites; and (4) deploy it as a tool that scans a real product page from its link.

2. Dataset

We used the public e-commerce dark pattern dataset of Yada et al. (2022), derived from the Mathur et al. crawl. It contains **2,356 text snippets** from real shopping websites, balanced between **1,178 dark pattern** and **1,178 normal** texts (e.g. "Write a review", "Size chart"). Dark pattern texts are labelled Scarcity (418), Social Proof (312), Urgency (210), Misdirection (195), Obstruction (27), Sneaking (12) or Forced Action (4); we merged the last three into **Other** because they have too few examples. The data was split 80/20 into **1,884 training** and **472 test** texts, stratified by category.

3. Methodology

3.1 TF-IDF baselines. Each text becomes a TF-IDF vector over unigrams and bigrams, and two linear classifiers are trained on it: **Logistic Regression** ($C = 10$) and a **Linear SVM** ($C = 1$).

3.2 Fine-tuned DistilBERT. DistilBERT (Sanh et al., 2019) is a compressed BERT that is 40% smaller while keeping about 97% of its language understanding. Unlike TF-IDF, which treats a text as a bag of words, it reads each word in context. We added a classification head to distilbert-base-uncased and fine-tuned all weights for 3 epochs (AdamW, learning rate $2e-5$ with linear decay, batch size 16, maximum 64 tokens), holding out 10% of the training set for validation.

3.3 Category classifier. A TF-IDF + Logistic Regression model with balanced class weights, trained only on dark pattern texts, names the type. In the full pipeline, DistilBERT first decides whether a line is a dark pattern (probability > 0.5); if it is, the category classifier labels it.

3.4 Domain adaptation for Indian websites. On Indian pages the model flagged harmless lines such as "Free delivery on orders above Rs. 499", rupee prices, ratings and product names. We generated **368 Indian-style training lines** from templates (rupee prices, delivery thresholds, ratings, product specifications, discount labels) plus dark patterns in the same style, so the model cannot simply learn that "₹" means safe. These were added to the training split only. For evaluation we wrote a separate **held-out set of 32 lines** (20 normal, 12 dark) with different wording.

4. Results

All models were evaluated on the same 472 test texts (Table 1).

Model	Accuracy	Precision	Recall	F1-score
TF-IDF + Logistic Regression	95.55%	94.61%	96.61%	95.60%
TF-IDF + Linear SVM	95.13%	94.19%	96.19%	95.18%
DistilBERT (fine-tuned)	96.82%	96.62%	97.03%	96.83%

Table 1: Dark pattern vs. normal classification on the test set (best model in bold).

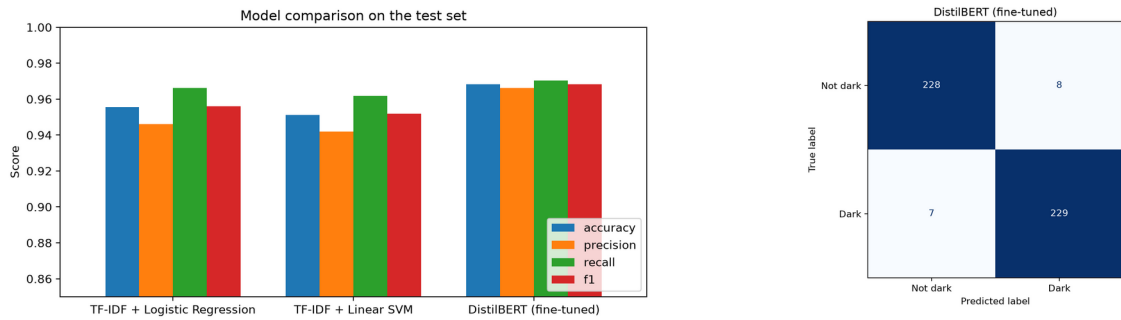


Figure 1: Model comparison (left) and DistilBERT confusion matrix on the test set (right).

DistilBERT performed best with 96.83% F1, making 15 mistakes on 472 test texts against 21 for Logistic Regression and 23 for the SVM. The TF-IDF baselines were still strong (about 95% F1) because many dark patterns use distinctive words; the highest-weighted were "left", "only", "purchased", "bought", "hurry", "limited time", "no thanks" and "ago". DistilBERT's advantage is context: "only" signals pressure in "only 3 left" but is harmless in "only available in cotton", which a bag-of-words model cannot distinguish. The **category classifier** reached 95.34% accuracy (macro F1 0.929); per-class F1 ranged from 0.982 (Scarcity) to 0.842 (Other, which has only 43 examples).

Domain adaptation (Table 2) raised accuracy on unseen Indian text sharply with no loss on the original test set. In our scans of the live Flipkart home page, flagged lines fell from 15 to 2. The held-out set is small, so these figures show a clear trend rather than a precise measurement.

DistilBERT model	Original test acc.	Original test F1	Indian held-out acc.	False alarms (of 20)
Before adaptation	97.03%	97.06%	65.63%	11
After adaptation	97.25%	97.24%	96.88%	1

Table 2: Effect of Indian-style training data. These models use the full training split (no validation hold-out), so their test scores differ slightly from Table 1.

5. System Implementation

The adapted model is deployed as a web application. Figure 2 shows how a scan flows through the system: the FastAPI server downloads the page, BeautifulSoup splits it into its visible lines of text in page order, and every line is classified in a single batch. When hosted publicly, the server refuses links that point to private network addresses and limits pages to 5 MB.

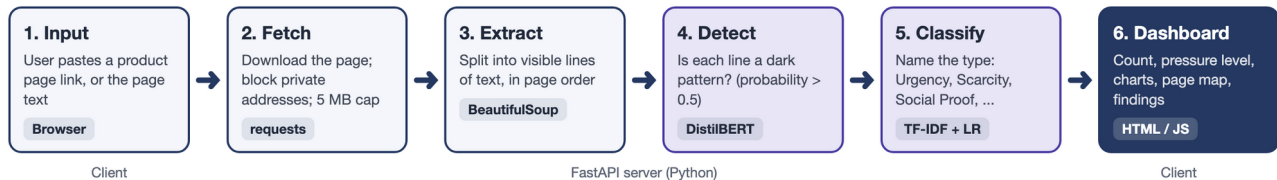


Figure 2: System architecture and data flow for one scan.

Component	Technology	Role
Model training	PyTorch, Hugging Face Transformers, scikit-learn	Fine-tune DistilBERT; train TF-IDF classifiers
Experiments	Jupyter notebook on Google Colab (T4 GPU)	Compare models, produce figures
Backend	FastAPI, requests, BeautifulSoup	Fetch pages, extract text, run the models
Frontend	HTML, CSS, JavaScript (no framework)	Dashboard, charts and scan animation
Packaging	Docker	Container for cloud hosting (e.g. Hugging Face Spaces)

Table 3: Technology stack.

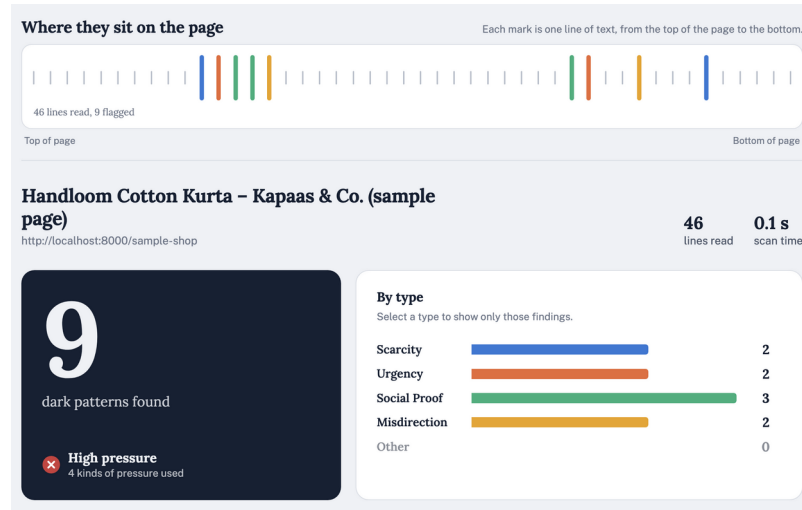


Figure 3: The dashboard after scanning a test product page that we built with dark patterns in it.

The dashboard (Figure 3) shows the number of dark patterns and a pressure level, a bar chart by category (click a bar to filter), a strip marking where each dark pattern sits on the page, and a list of findings with the model's confidence and the closest CCPA 2023 pattern. Users can also paste page text directly for sites that block automated access. **To run it:**

```
pip install -r app/requirements.txt
python app/train.py          # trains and saves the model (about 3 minutes on a laptop GPU)
python app/server.py        # then open http://localhost:8000
```

6. Limitations and Future Work

- **Text only:** visual tricks such as pre-ticked checkboxes, tiny "decline" buttons or hidden costs need a vision model trained on page screenshots.
- **Language, not truth:** the model detects manipulative phrasing; it cannot tell whether "Only 2 left" is actually false.
- **Regional data:** our Indian data is template-generated and the held-out set is small; short generic lines such as "In stock" are still sometimes flagged. A larger hand-labelled set from real Indian sites, including Hinglish, is the next step.
- **Page coverage:** sites that block bots or build content with JavaScript cannot be scanned by link. A browser extension would read the page exactly as the user sees it.

7. Conclusion

We built an end-to-end NLP system that detects dark patterns in e-commerce text. Fine-tuned DistilBERT reached 96.82% accuracy, outperforming TF-IDF baselines, and a second classifier names the type of manipulation with 95.34% accuracy. Domain adaptation raised accuracy on Indian text from 65.63% to 96.88%, and the model now powers a dashboard that scans real product pages. Tools like this can help consumers spot manipulation and help regulators such as the CCPA enforce the 2023 guidelines at scale.

References

- [1] Mathur, A., Acar, G., Friedman, M. J., Lucherini, E., Mayer, J., Chetty, M., & Narayanan, A. (2019). Dark Patterns at Scale: Findings from a Crawl of 11K Shopping Websites. Proceedings of the ACM on Human-Computer Interaction (CSCW).
- [2] Yada, Y., Feng, J., Matsumoto, T., Fukushima, N., Kido, F., & Yamana, H. (2022). Dark Patterns in E-Commerce: A Dataset and Its Baseline Evaluations. IEEE International Conference on Big Data.
- [3] Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. arXiv:1910.01108.
- [4] Central Consumer Protection Authority, Government of India (2023). Guidelines for Prevention and Regulation of Dark Patterns, 2023.

Source code: <https://github.com/ketanabhishek8/Dark-Pattern-Website-Detector>